

Chromatin regulators mediate anthracycline sensitivity in breast cancer

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Anthracyclines are a highly effective component of curative breast cancer chemotherapy but are associated with substantial morbidity^{1,2}. Because anthracyclines work in part by inhibiting topoisomerase-II (TOP2) on accessible DNA^{3,4}, we hypothesized that chromatin regulatory genes (CRGs) that mediate DNA accessibility might predict anthracycline response. We studied the role of CRGs in anthracycline sensitivity in breast cancer through integrative analysis of patient and cell line data. We identified a consensus set of 38 CRGs associated with anthracycline response across ten cell line datasets. By evaluating the interaction between expression and treatment in predicting survival in a metacohort of 1006 patients with early-stage breast cancer, we identified 54 CRGs whose expression levels dictate anthracycline benefit across the clinical subgroups; of these CRGs, 12 overlapped with those identified in vitro. CRGs that promote DNA accessibility, including Trithorax complex members, were associated with anthracycline sensitivity when highly expressed, whereas CRGs that reduce accessibility, such as Polycomb complex proteins, were associated with decreased anthracycline sensitivity. We show that *KDM4B* modulates TOP2 accessibility to chromatin, elucidating a mechanism of TOP2 inhibitor sensitivity. These findings indicate that CRGs mediate anthracycline benefit by altering DNA accessibility, with implications for the stratification of patients with breast cancer and treatment decision making.

We compiled a list of 404 CRGs from literature review and Gene Ontology (GO) annotation (Supplementary Table 1 and Methods). To evaluate the association between the mRNA expression of these CRGs and anthracycline response in human breast cancers, we combined data from multiple sources, including The Cancer Genome Atlas (TCGA) breast cancer cohort⁵, breast cancer cell line expression and drug response data^{6–13} and a clinically annotated collection of expression profiles for 1,006 patients with early-stage breast cancer (Fig. 1a and Methods). We focused on CRG mRNA expression because CRGs are infrequently mutated in breast cancer but their copy number is often altered, presumably effecting expression changes (Extended Data Fig. 1a)^{14,15}. We evaluated whether CRGs have a central regulatory role in breast cancer using graph theoretical approaches. First, we generated a transcriptional regulatory network from TCGA breast tumor RNA-seq data ($n=1,079$ patients) using the algorithm for the reconstruction of accurate cellular networks (ARACNE)¹⁶, assuming each gene is a regulatory element.

The set of CRGs exhibited high network centrality¹⁷ as measured by degree (3.26 ± 4.37 in CRGs versus 2.04 ± 3.7 in non-CRGs), and this was significantly greater ($P < 1 \times 10^{-4}$) than for a null distribution, with similar results for betweenness and page rank (Extended Data Fig. 1b,c and Methods). CRGs were also significantly enriched for influencers¹⁸ (Fisher's exact one-tailed test, $P < 9 \times 10^{-23}$; odds ratio (OR) = 2.68) (Extended Data Fig. 1d). To identify the sets of target genes directly regulated by each CRG, we used ARACNE to generate a breast cancer chromatin regulatory network, where CRGs correspond to hubs and target genes are terminal nodes (Fig. 1a and Extended Data Fig. 2a).

We then defined a genome-wide signature of anthracycline response by analyzing ten breast cancer cell line datasets with expression data (RNA-seq or microarray) and dose-response metrics (GI_{50} and AUC)^{6–13} (Fig. 1a), where the F statistic (per gene) was used as a measure of treatment response. For each dataset, we identified a signature of anthracycline response by performing differential expression analysis between cell lines that were resistant or sensitive to doxorubicin (Extended Data Fig. 2b,c, Supplementary Table 2 and Methods). We next used virtual inference of protein-activity by enriched regulon (VIPER)¹⁹ analysis to identify genes from the breast cancer chromatin regulatory network whose putative targets were significantly enriched in the anthracycline response signature. This yielded a consensus list of 38 CRGs that were associated (Fisher's combined $P < 0.05$) with anthracycline response across multiple in vitro datasets (Fig. 1b,c).

We next evaluated the association between the 404 CRGs and anthracycline benefit in a meta-cohort of 1,006 patients with early-stage breast cancer for whom tumor characteristics, overall survival, treatment and gene expression data were available^{20–25} (Fig. 1, Extended Data Fig. 3, Supplementary Table 3 and Methods). We used a Cox proportional hazards model to study the interaction between gene expression and treatment and their association with overall survival²⁶. The model was adjusted by clinical covariates known to be associated with breast cancer outcome—including estrogen receptor (ER), progesterone receptor (PR) and HER2 status, tumor size (T stage), MKI67 expression and lymph node status—as well as by cohort. We first compared patients treated with anthracyclines ($n=218$) to those who were not (including those who were treated with other chemotherapies or endocrine therapy alone or who received no treatment) ($n=542$)²⁶. We found 54 CRGs with an interaction between CRG expression and treatment (anthracycline versus no anthracycline) that was associated ($P < 0.05$) with

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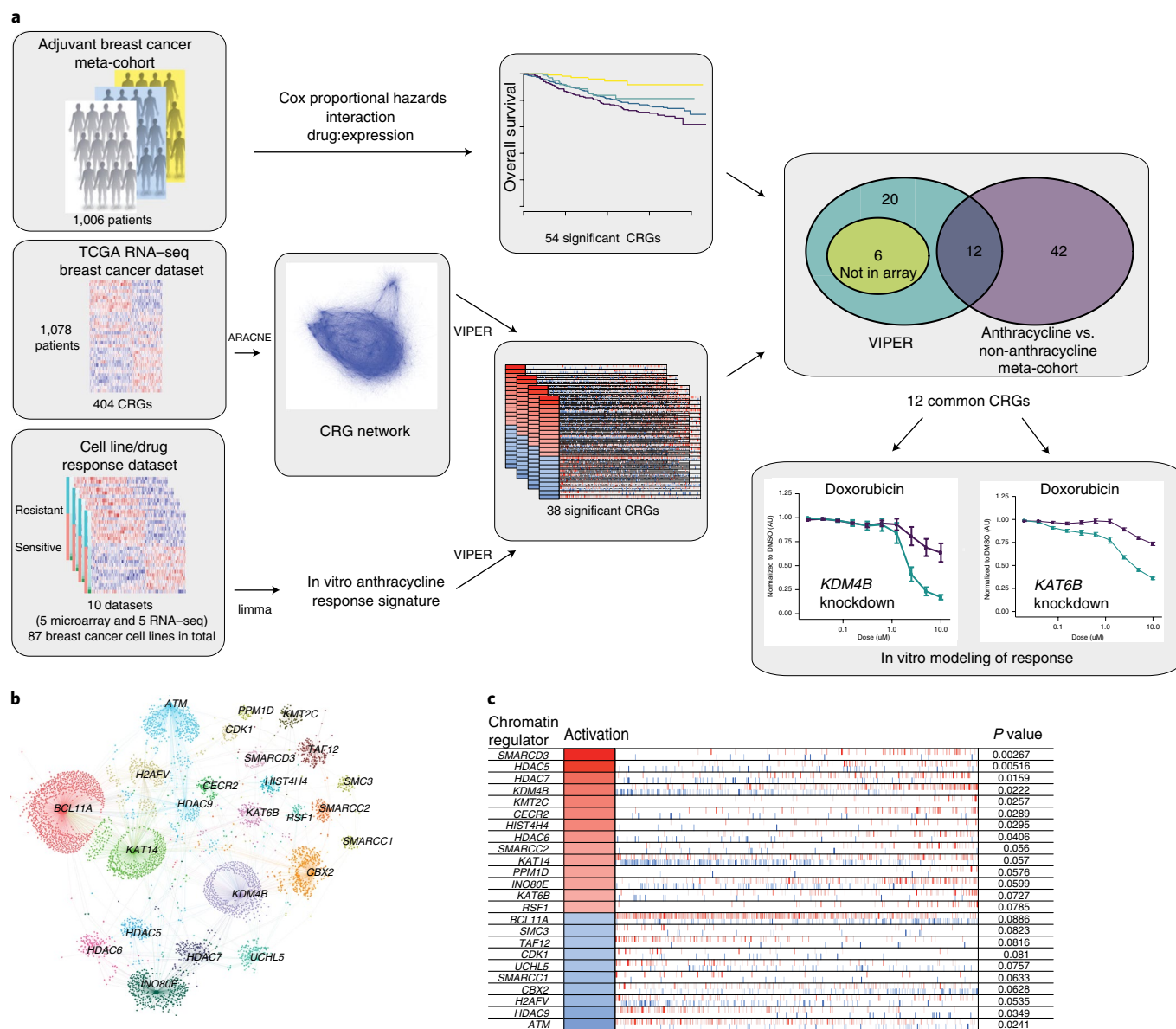


Fig. 1 | Schematic overview of study design, the analytical and experimental framework and derivation of an anthracycline response signature. a, An adjuvant breast cancer metacohort of 1006 clinically annotated patients with early-stage breast cancer and gene expression data was used to identify genes for which the interaction between expression levels and treatment with anthracyclines was significantly associated with outcome, resulting in 54 CRGs. RNA-seq data from the TCGA breast cancer cohort were used to infer a breast cancer CRG network using the ARACNE algorithm. A panel of 87 breast cancer cell lines from 10 datasets with accompanying gene expression data and dose-response curve metrics (GI_{50} or AUC) data was used to build a genome-wide signature of anthracycline response (Methods). On the basis of the dose-response curve, cells were classified as sensitive or resistant (upper and lower third of the dose-response values, respectively). The VIPER algorithm was then applied to each dataset to identify CRGs from the ARACNE network whose targets were significantly enriched in the anthracycline response signature, yielding a consensus list of 38 CRGs. The intersection of genes that were significant in both the in vitro and in vivo analyses yielded 12 CRGs, from which *KDM4B* and *KAT6B* were selected for functional evaluation. **b**, CRGs enriched in the in vitro doxorubicin signature based on VIPER are shown for the Heiser microarray dataset ($n = 46$ biologically independent cell lines). CRGs are labeled within the network, and the corresponding target genes whose expression they modify are indicated as individual dots. **c**, For each CRG associated with anthracycline response in vitro, the inferred activation status is indicated (red, positive association with anthracycline sensitivity; blue, negative association with anthracycline sensitivity), as is the change in expression of its target genes, as indicated by individual bars along the horizontal axis (red, upregulated; blue, downregulated), and the significance of the association (P value based on comparison of the enrichment score with a null distribution generated from 1000 random signatures, not adjusted). The order of target genes is defined by the strength of association with doxorubicin response.

overall survival (Extended Data Fig. 4 and Supplementary Table 4). There was positive enrichment of gene-drug interactions associated ($P < 0.05$ from the Cox proportional hazards model) with outcome among CRGs (Fisher's exact one-tailed test, $P = 0.00062$, OR = 1.54; Extended Data Fig. 5). Notably, a subset of CRGs was associated

with reduced anthracycline benefit when their expression levels were below the median, and many of these genes promote open chromatin. This list included Trithorax-group proteins, including the BAF complex subunits *ARID1A*, *SMARCD3*, *SMARCD1* and *SMARCA2*; COMPASS complex subunits such as *KMT2A*; and the

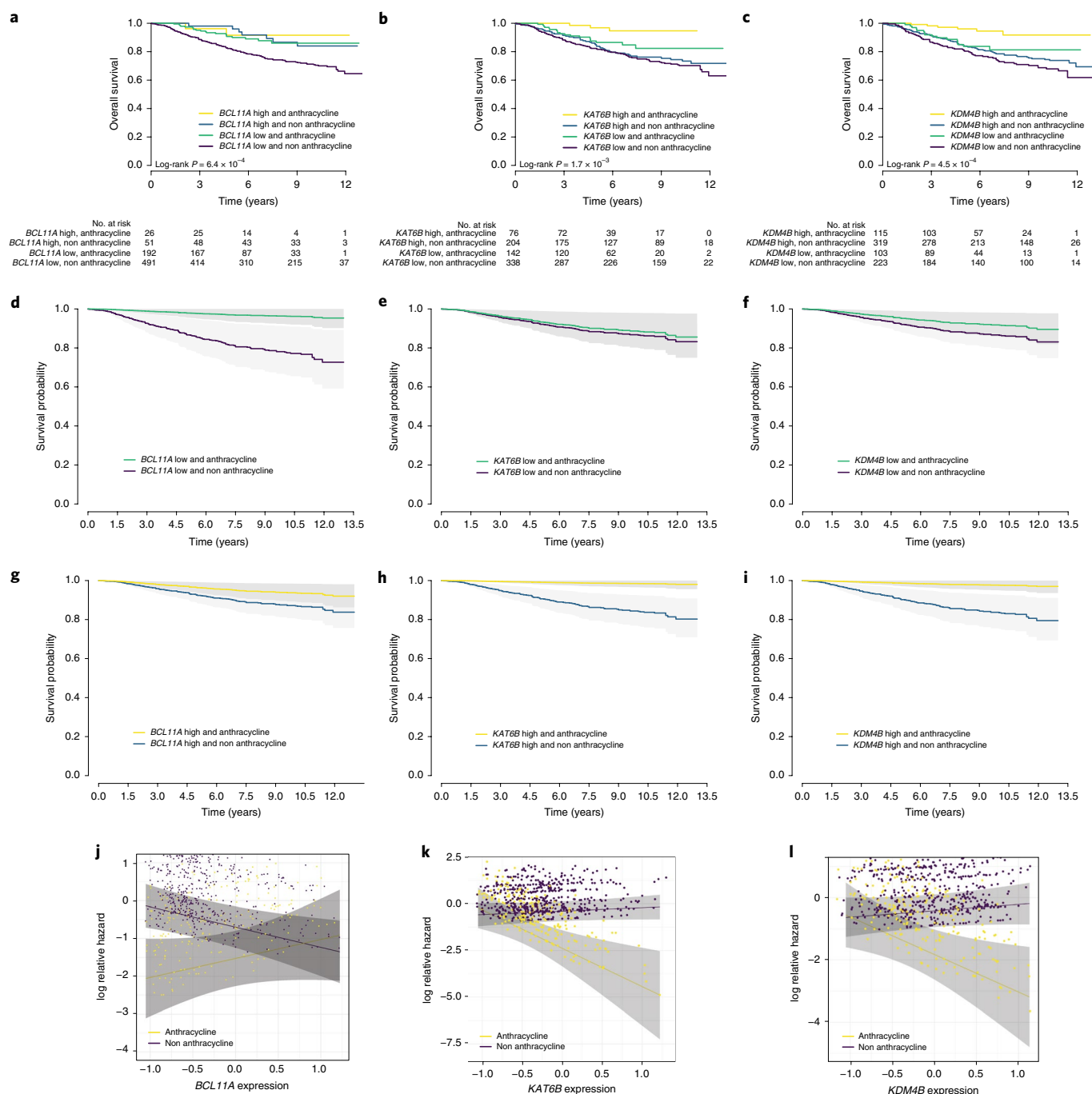


Fig. 2 | The expression levels of key chromatin regulatory genes are associated with anthracycline response. **a–c**, Kaplan–Meier plots of anthracycline-treated and non-treated patients ($n = 760$ patients) stratified by high versus low expression of key CRGs: *BCL11A* (**a**), *KAT6B* (**b**) and *KDM4B* (**c**). P values from a two-sided log-rank test. **d–f**, Cox proportional hazards models of the probability of overall survival (adjusted by hormone receptor status, HER2 status, lymph node status, T stage, MKI67 expression and cohort) for low expression (expression greater than 1 s.d. below the median) of *BCL11A* (**d**), *KAT6B* (**e**) and *KDM4B* (**f**). Gray shading corresponds to 95% confidence intervals. **g–i**, As in d–f but for high expression (expression greater than 1 s.d. above the median) of *BCL11A* (**g**), *KAT6B* (**h**) and *KDM4B* (**i**) in treated versus untreated samples. Notably, this corresponds to a threshold-free representation of the hazards. Gray shading corresponds to 95% confidence intervals. Points represent patients.

histone lysine acetyltransferase *KAT6B* and histone demethylases *KDM6B* and *KDM4B*. Conversely, a separate subset of CRGs was associated with greater anthracycline benefit when their expression levels were below the median. This included the Polycomb gene *EZH2* as well as the histone deacetylase *HDAC9* and the histone chaperones *RSF1* and *BCL11A*, whose role in chromatin accessibility is less clear; examples of both patterns are provided in Fig. 2.

Collectively, these findings indicate that CRGs predict anthracycline benefit in patients with early-stage breast cancer.

The observation that higher expression of BAF complex subunits or lower expression of Polycomb subunits is associated with anthracycline sensitivity fits with previous models³. TOP2 proteins function as dimers of approximately 340 kDa (nearly twice the size of a nucleosome) that require accessible chromatin to bind DNA.

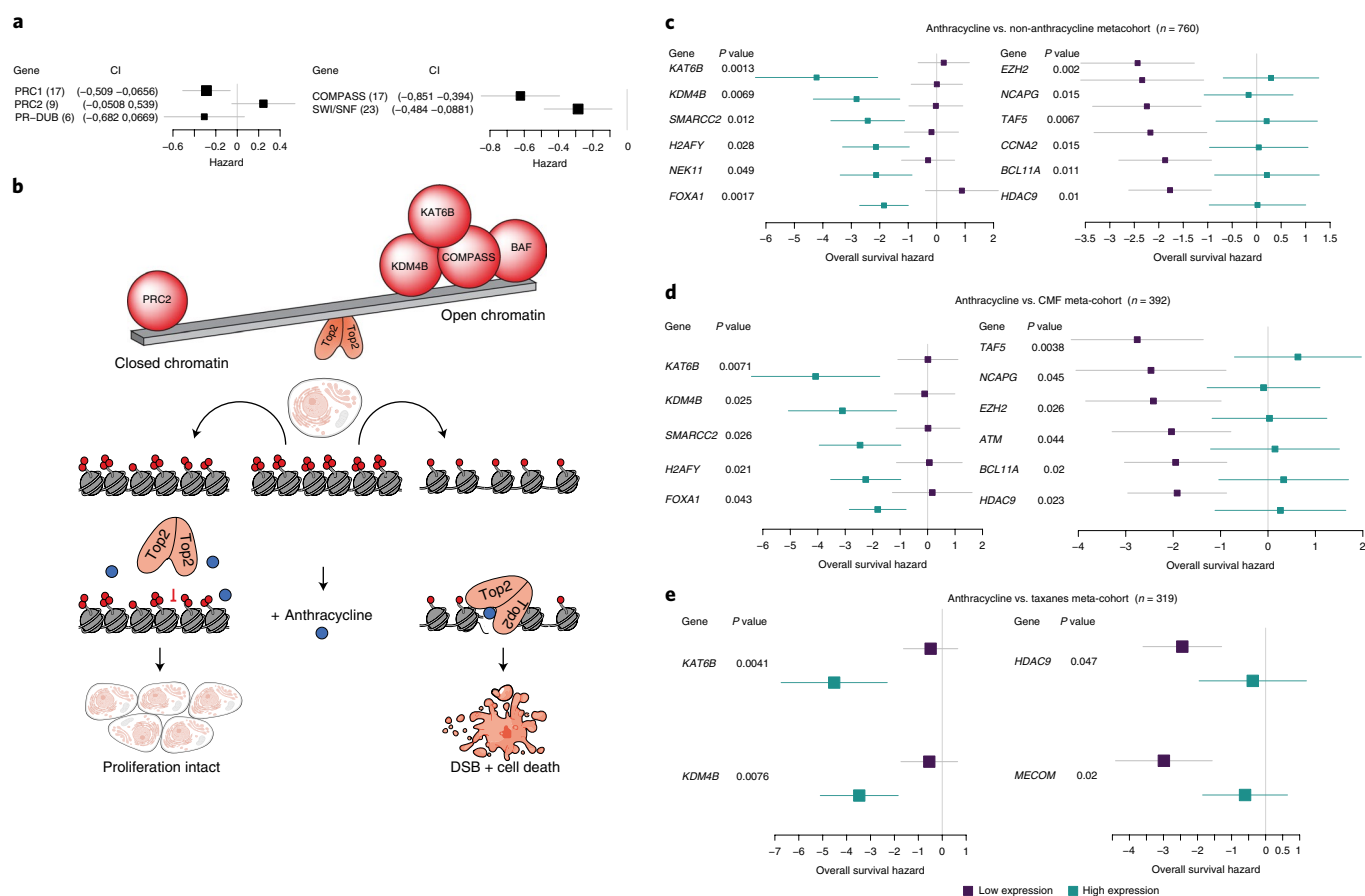


Fig. 3 | Chromatin state mediates anthracycline response. **a**, High expression levels of Trithorax complexes (COMPASS and SWI/SNF) are associated with better outcome in anthracycline-treated patients ($n=760$ patients), whereas high expression of the PRC2 complex is associated with worse outcome; the hazard ratio plots summarize the results for genes in each complex across the meta-cohort. The number of genes in each complex is indicated in parentheses. **b**, Schematic illustration of a model wherein the Trithorax and PRC2 genes have opposing roles in mediating anthracycline response. In this scenario, BAF and COMPASS complex subunits, KDM4B, KAT6B and others open the DNA fiber to TOP2 binding, thereby increasing anthracycline sensitivity, accompanied by double-strand breaks (DSB) and cell death, whereas an opposing set of CRGs—including Polycomb proteins (PRC2), BCL11A, HDAC9 and others—close the DNA fiber to TOP2 binding, thereby decreasing anthracycline sensitivity such that proliferation remains intact. **c–e**, Forest plots depicting the hazards of overall survival based on the expression of CRGs that were also associated with anthracycline response in vitro (from VIPER). P values are based on the Cox proportional hazards model of the interaction between drug and gene expression, adjusted by hormone receptor status, HER2 status, T stage, lymph node status, *MKI67* expression from the expression array data and cohort (Methods). The forest plots contrast treated versus non-treated patients in two settings, namely high CRG expression (cyan) and low CRG expression (purple). Left, representative genes for which high expression is associated with better outcome (negative hazards) in patients treated with anthracyclines; right, representative genes for which low expression is associated with better survival in patients treated with anthracyclines. Varied chemotherapy regimens are compared, namely anthracycline-based treatment versus non-anthracycline-based treatment (CMF, taxanes, hormonal therapy or no treatment; $n=760$ patients) (**c**) or subsets (anthracycline versus CMF, $n=392$ patients (**d**); anthracycline versus taxanes, $n=319$ patients (**e**)). The forest plots show the $\log_2$ -transformed hazard ratios (box), and error bars represent 95% confidence intervals.

A functional BAF complex is necessary for TOP2 to associate with DNA at about 60% of its sites in the genome. Loss of TOP2 function in cells with a dysfunctional BAF complex renders cells less sensitive to TOP2 inhibitors³. Conversely, the Polycomb complex antagonizes the BAF complex^{27,28}, conferring TOP2 inhibitor resistance.

Because the BAF complex, a member of the Trithorax group, influences TOP2 recruitment and accessibility and opposes Polycomb-group complexes²⁹, we evaluated the roles of these two complexes in mediating anthracycline benefit. Summarizing the hazard ratios for all genes in each complex across the breast cancer meta-cohort, we found that in the presence of anthracyclines higher expression of Polycomb repressive complex 2 (PRC2) genes was generally associated with a higher hazard ratio, whereas higher expression of both BAF and COMPASS, members of the Trithorax class of genes, was generally associated with lower hazard ratios (Fig. 3a and Extended Data Fig. 6). Recent studies showed

that changes in PRC1 levels did not lead to concomitant changes in accessibility, which was consistent with the lack of a change in the hazard ratio for PRC1 or PR-DUB genes (Extended Data Fig. 6)^{30,31}. Thus, those CRGs for which high expression was associated with greater anthracycline benefit were generally associated with increased DNA accessibility, whereas those for which high expression was associated with less anthracycline benefit were associated with decreased DNA accessibility. The Trithorax proteins, including BAF and COMPASS complexes, KDM4B, KAT6B and others, open the DNA fiber for TOP2 binding, thereby increasing anthracycline sensitivity. Conversely, an opposing set of CRGs, including Polycomb-group proteins (PRC2 complex) and others, close the DNA fiber to TOP2 binding, thereby decreasing anthracycline sensitivity (Fig. 3b). These data are consistent with a model wherein the expression levels of Trithorax and Polycomb genes act in opposing directions to mediate anthracycline benefit.

To assess whether the CRGs associated with anthracycline benefit were also more generally implicated in benefit to other chemotherapies, we performed several additional comparisons. First, we repeated these analyses comparing patients treated with anthracyclines ($n=218$) and patients treated with the chemotherapy regimen cyclophosphamide/methotrexate/5-fluorouracil (CMF), which does not contain an anthracycline ($n=174$). Here we identified 44 CRGs with a significant ($P<0.05$) interaction between expression and treatment in predicting overall survival (Supplementary Table 5). Second, we repeated these analyses comparing patients treated with anthracyclines and no taxanes ($n=196$) to patients treated with taxanes and no anthracyclines ($n=123$). Here we identified 50 CRGs with a significant ($P<0.05$) interaction between their expression and treatment in predicting overall survival (Supplementary Table 6).

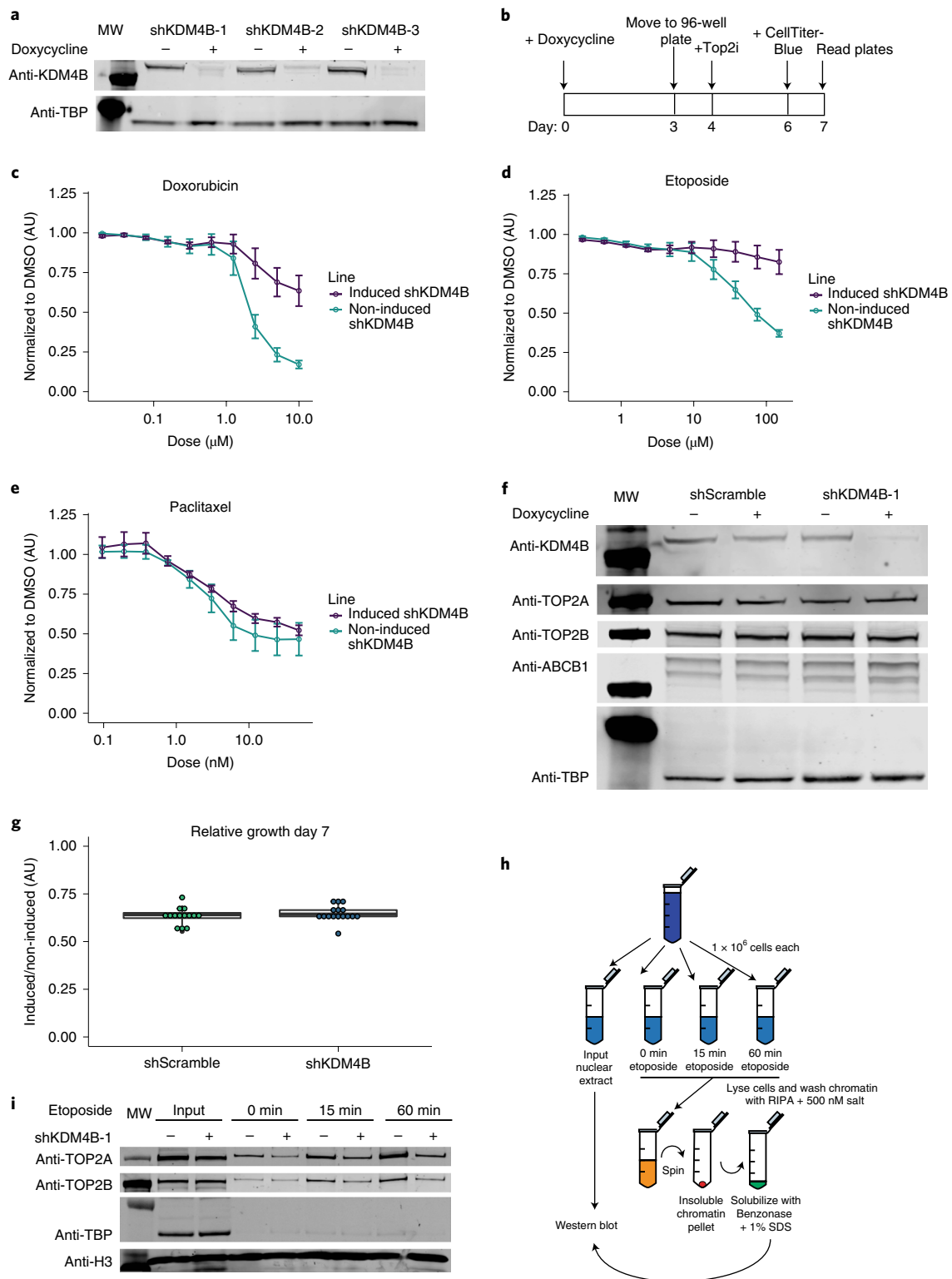
Of the 38 CRGs implicated in anthracycline response in vitro, 32 had available expression data in the meta-cohort; of these, 12 exhibited a significant interaction between expression and anthracycline usage in predicting overall survival when comparing anthracycline-treated and non-anthracycline-treated patients (Fig. 3c and Extended Data Fig. 7a). As expected, the VIPER enrichment factors for CRGs were negatively correlated (Pearson's $R=-0.77$, $P=3.09 \times 10^{-3}$) with the hazard ratio in the meta-cohort (Extended Data Fig. 7d). Among the 44 CRGs that were significant when comparing anthracycline-treated versus CMF-treated patients, 11 genes were also significant in the in vitro analysis (Fig. 3d and Extended Data Fig. 7b). Of the 50 genes from the anthracycline-treated versus taxane-treated comparison, 4 were significant in the in vitro analysis (Fig. 3f and Extended Data Fig. 7c). Among the 22 CRGs shared across the 3 treatment comparisons in the patient metacohort (Extended Data Fig. 7e), three (*KDM4B*, *KAT6B* and *HDAC9*) were also significant in vitro. We further assessed anthracycline specificity by comparing enrichment of the 12 genes common across the in vitro and in vivo analyses with the VIPER enrichment for other breast cancer chemotherapies (Supplementary Fig. 1). Doxorubicin was generally the most enriched, but docetaxel exhibited modest enrichment, albeit at a lower level. Collectively, these results suggest that the CRGs identified through these analyses are implicated in anthracycline sensitivity rather than general chemosensitivity.

While our initial analyses adjusted for ER, PR and HER2 status, we further sought to determine whether the associations remained significant within each of the clinical subgroups rather than being driven by one subtype. We therefore stratified the meta-cohort into the three clinical subtypes: ER-positive/HER2-negative breast cancer ($n=204$; Supplementary Table 7), HER2-positive breast cancer ($n=216$; Supplementary Table 8) and triple-negative breast cancer

(TNBC) ($n=113$; Supplementary Table 9). For the ER-positive/HER2-negative group, we included hormone therapy as a covariate. Notably, across these three patient subgroups, the directionality of the hazard ratios for most of the 54 CRGs that were significant in the full cohort remained the same, despite the reduction in sample size in these subset analyses (Extended Data Fig. 8 and Methods). Examination of CRG expression by subtype revealed higher expression of *KDM4B* and *KAT6B* and lower expression of *BCL11A* in ER-positive versus ER-negative tumors (Supplementary Fig. 2). Although ESR1 signaling was correlated with CRG expression in both subgroups, substituting this for ER status in the model did not change the results. Collectively, these findings indicate that CRG expression is associated with anthracycline benefit across the breast cancer clinical subgroups.

In both the cell line and patient data, reduced *KDM4B* expression was strongly associated with anthracycline resistance. In vitro, *KDM4B* is a weak histone demethylase that recognizes H3K9me2 and H3K9me3 and converts the histone tail to H3K9me1, effectively changing the histone mark to one that is associated with a more accessible, transcriptionally active chromatin state. To functionally evaluate the role of *KDM4B* in anthracycline response, we searched for a cell line that could serve as a model to recapitulate the phenomenon seen in patients and found the HCC1954 breast cancer cell line to be suitable (data for other cell lines that failed to recapitulate the patient data can be found in Supplementary Fig. 3). Levels of *KDM4B* protein were lowered using three inducible short hairpin RNA (shRNA) knockdown constructs in HCC1954 cells (Fig. 4a). After *KDM4B* knockdown for 4 d, cells were treated with doxorubicin, etoposide (a non-anthracycline TOP2 inhibitor) or paclitaxel (a taxane that functions via tubulin inhibition) for 3 d, after which cell viability was measured (Fig. 4b). Consistent with the patient data, where CRG expression levels predicted response to anthracycline but not taxane treatment, *KDM4B* knockdown induced resistance to doxorubicin (Fig. 4c) and etoposide (Fig. 4d), but cells remained sensitive to paclitaxel (Fig. 4e). An inducible scrambled shRNA did not result in significant changes in sensitivity to drug treatment (Extended Data Fig. 9a–c). We confirmed that knockdown-induced anthracycline resistance was not due to decreases in cell proliferation, loss of the drug target (TOP2A or TOP2B) or upregulation of the ABCB1 multidrug exporter protein (Fig. 4f,g). In the patient meta-cohort, there was also minimal ($|R|<0.2$) correlation between *KDM4B* and *TOP2A*, *TOP2B* or *ABCB1* expression (Extended Data Fig. 9e). Knockdown of *KAT6B* similarly resulted in anthracycline resistance in HCC1954 cells, consistent with the patient data and the coordinate role of multiple CRGs in anthracycline response (Extended Data Fig. 10).

Fig. 4 | *KDM4B* mediates TOP2 inhibitor response in breast cancer cells by modulating TOP2 accessibility to chromatin. **a**, Western blots on nuclear extracts showing inducible shRNA knockdown of *KDM4B* with three unique shRNA sequences in the HCC1954 breast cancer cell line. See Source Data Fig. 4. Experiments were repeated independently three times with similar results. **b**, Schematic overview of drug response assays. **c**, Doxorubicin dose-response curves for *KDM4B*-induced shRNAs (shRNA 1, 2, 3) in HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=9$; three independent experiments of three independent shRNAs each. Center value is mean $\pm$ s.e.m. **d**, Etoposide dose-response curve for *KDM4B*-induced shRNAs (shRNA 1, 2, 3) in HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=9$; three independent experiments of three independent shRNAs each. Center value is mean $\pm$ s.e.m. **e**, Paclitaxel dose-response curve for *KDM4B*-induced shRNAs (shRNA 1, 2) in HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=3$; three independent experiments from two independent shRNAs. Center value is mean $\pm$ s.e.m. **f**, Western blot on whole-cell extract showing knockdown with a scramble shRNA or knockdown of *KDM4B* (shRNA-1) with no concomitant loss of drug targets TOP2A and TOP2B or a gain in the drug efflux pump ABCB1. See Source Data Fig. 4; subfigure is a combination of multiple blots. Experiments were repeated independently three times with similar results. **g**, Relative cell number of HCC1954 cells with induced shRNAs to non-induced cells (shRNAs to *KDM4B* 1, 2, 3 and shRNA Scramble) at day 7 (induced/DMSO vehicle). $xn=14$, Scramble, and $n=16$, *KDM4B*, from independent experiments. Center line of box is the median data point with upper bound of box the third quartile and lower bound of box the first quartile. Whiskers represent $1.5 \times$ the IQR. **h**, Schematic of etoposide fixed, high salt wash TOP2 DNA accessibility experiments shown in **i** and Extended Data Fig. 9. **i**, Western blot of input samples and resolubilized chromatin pellet after wash with 500 mM NaCl from HCC1954 cells treated with etoposide for given time, $\pm$ induction of shRNAs shows a reduction of chromatin-bound TOP2 proteins in the absence of *KDM4B*. Histone H3 is shown as a loading control, and TBP shows removal of transiently bound proteins by 500 mM salt wash. See Source Data Fig. 4. Experiments were repeated independently three times with similar results.



Next, we evaluated whether KDM4B modulated TOP2 binding to chromatin and, therefore, sensitivity to anthracyclines and other TOP2 inhibitors. Because etoposide treatment leads to a covalent cross-link between TOP2 proteins and DNA that can occur only where TOP2 proteins have access to DNA, this provides a powerful way to determine TOP2 accessibility to chromatin^{3,4} (Fig. 4h). *KDM4B* knockdown in HCC1954 cells, followed by etoposide treatment for 15 or 60 min and a 500 mM salt wash to remove transiently bound proteins, resulted in less TOP2A and TOP2B bound to the

insoluble chromatin fraction (Fig. 4i), which was not observed in cells induced to express the scrambled control shRNA (Extended Data Fig. 9d). Thus, we showed that resistance to TOP2 inhibitors after knockdown of *KDM4B* is due to a direct loss of TOP2 accessibility to chromatin, highlighting the specificity of CRGs in anthracycline response.

In summary, we identified a network of CRGs and showed that naturally occurring variations in their expression levels dictate chromatin accessibility and sensitivity to anthracycline therapy across

clinically defined breast cancer subtypes. We found that CRGs that tend to increase DNA accessibility were generally associated with greater anthracycline benefit when expressed at high levels, whereas CRGs associated with decreased DNA accessibility tended to be associated with anthracycline resistance when expressed at high levels. These results are consistent with a model in which CRGs including the BAF and COMPASS complexes, KDM4B and others make DNA available for TOP2 binding, thereby increasing anthracycline sensitivity, whereas an opposing set of CRGs, including PRC2, reduces DNA availability to TOP2 binding, thereby decreasing anthracycline sensitivity.

Our results have several implications. There is an urgent need to understand the biology that determines anthracycline response, especially in early-stage breast cancer, where the modestly more efficacious anthracycline-containing regimens³² are associated with clinically important risks of cardiac toxicity and secondary malignancy^{1,2}. Although commercially available gene expression tests can be used to help determine whether a patient with early-stage breast cancer is likely to benefit from chemotherapy^{33–36}, there are presently no biomarkers that guide the use of anthracycline- versus non-anthracycline-containing regimens. Notably, we showed that chromatin state is associated with anthracycline benefit in ER-negative as well as ER-positive disease, and in node-negative as well as node-positive breast cancer. Therefore, a signature based on the expression levels of the CRGs defined here could be used to distinguish patients with early-stage breast cancer who will benefit from an anthracycline-containing regimen from those who will not. Further evaluation of such a predictive signature is warranted before its implementation as a diagnostic assay. Such an approach has the potential to inform patient stratification, resulting in improved outcomes while mitigating toxicity. The chromatin accessibility biology elucidated here may also guide approaches to modify CRG function to improve anthracycline sensitivity in patients who truly need it, such as with histone deacetylase inhibitors and other epigenetically targeted therapies.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-019-0638-5>.

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Methods

Breast cancer cell line and patient datasets. The TCGA breast cancer RNA-seq dataset ($n = 1,079$ patients) was downloaded from <https://gdc.cancer.gov> in January 2018. RPKM count data were normalized using variance stabilizing transformation (VST) from the DESeq2 package³⁷ within R Bioconductor. The breast cancer cell line drug response datasets, including RNA-seq, gene expression microarray data and drug response information, were downloaded for the Heiser⁶ (RNA-seq, microarray and drug response) and Gray⁸ (drug response) datasets and using the PharmagoGX package³⁸ (gCSI, RNA-seq and drug response; GDSC1000, microarray and drug response; CCLE, RNA-seq and microarray; FIMM and CTRPv2, drug response). We used data from a total of 87 breast cancer cell lines that exhibited sensitivity or resistance to doxorubicin or other breast cancer chemotherapies. Drug response information was recorded for the Heiser dataset as $-\log_{10}(GI_{50})$, where GI_{50} was the concentration that inhibited cell growth by 50% after 72 h of treatment, and as the AUC for the rest of the datasets (area under the dose-response curve). Briefly, for each dataset, we divided the total number of cell lines into those that were sensitive (top tertile of $-\log_{10}(GI_{50})$ or AUC for the drug) or resistant (bottom tertile of $-\log_{10}(GI_{50})$ or AUC for the drug). We used limma³⁹ with weighted samples (arrayWeight function) for the microarray datasets and voom normalization for the RNA-seq datasets to obtain both a signature of response to anthracycline or other drugs and a null model for each signature by permuting the sample labels 1,000 times. P values and t statistics were based on limma (moderated t statistic).

Compilation of a meta-cohort of patients with breast cancer treated with adjuvant. Raw CEL files were downloaded from the Gene Expression Omnibus (GEO) database for the datasets from Kao et al.²² (GSE20685), IRB/JNR/NUH²¹ (GSE45255), MAIRE²⁵ (GSE65194), UPS²³ (GSE3494) and STK²⁴ (GSE1456). These datasets were each profiled on the Affymetrix platform (hgu133plus2, hgu133a and hgu133b), reprocessed using the rma function from the affy package⁴⁰ and quantile normalized. COMBAT⁴¹ was used to remove batch effects. Patients who received an anthracycline (doxorubicin or epirubicin) as a component of their treatment regimen were classified as 'anthracycline treated', whereas patients who received a chemotherapy regimen that did not contain anthracyclines, who received endocrine therapy alone or who received no therapy were classified as 'not anthracycline treated'. ER, PR and HER2 status was inferred using a Gaussian mixture model of probes 205225_at, 208305_at and 216836_s_t, respectively. MKI67 values were obtained from probe 212023_s_at. Lymph node positivity was a binary feature obtained from number of nodes > 0 or N stage ≥ 1 . T stage was a factor obtained from either the actual T stage, as reported ($n = 327$ cases), or as inferred from the reported size of the tumor ($T1 < 2$ cm, $T2 \leq 5$ cm and $T3 > 5$ cm) ($n = 520$ cases). For the STK cohort, information on T stage, lymph node status and N stage was not available; however, the authors reported that the mean tumor size in the cohort was 22 mm and 62% of samples were < 21 mm where 38% of samples were lymph node positive. We inferred T-stage of 2 and negative lymph node status for all samples in this cohort. The ESR1 signature was calculated using the list of genes from ref.⁴² and the function sig.score from the genu package⁴³.

Defining a set of chromatin regulators. We generated a list of chromatin regulators ($n = 404$; Supplementary Table 1) based on a defined set of GO functions, including (1) histone lysine methyltransferase activity (GO:0018024); (2) histone demethylation (GO:0032452); (3) histone deacetylation (GO:0004407); (4) histone acetyltransferase activity (GO:0004402); (5) histone phosphorylation (GO:0016572); (6) PRC1 complex (GO:0035102); (7) PRC2 complex (GO:0035098); (8) SWI/SNF complex (GO:0016514 plus other members not included in this GO category); (9) ISWI complex members (NURF, ACG, CHRAC, WICH, NORC, RSF) and CERF complex members^{44,45}; (10) chromodomain and NURD Mi-2 complex⁴⁶; (11) INO80 complex (GO:0031011); (12) SWR1 complex⁴⁷; (13) PR-DUB complex⁴⁸; (14) CAF1 complex (GO:0033186); (15) cohesins⁴⁹; (16) condensins⁵⁰; (17) topoisomerases (GO:0003916); and (18) DNA methyltransferases (GO:0006306), DNA demethylases (GO:0080111), histone proteins⁵¹ and chromatin pioneer factors⁵².

Network inference. From VST normalized gene expression data, we used the ARACNE-AP algorithm⁵³ to generate a genome-wide breast cancer transcriptional network (24,919 $\times$ 24,919 expression values) and CRG network (404 $\times$ 24,919 expression values). ARACNE-AP was run with the recommended parameters ($P < 1 \times 10^{-8}$). Significant networks were calculated from 10 bootstrap iterations for the genome-wide network and from 100 bootstraps for the CRG network. Network edges with adjusted P values > 0.05 were removed. The regulon was composed of 396 CRGs (8 had no nodes directly connected and, hence, were excluded), and the median number of targets per CRG was 94. We calculated the degree, betweenness and page rank centrality for each gene (node) in the genome-wide network using the igraph package (<http://igraph.org/r>). To compute a null distribution, we randomly selected 10,000 combinations of 404 genes (out of 24,919) and obtained centrality measures by aggregating the values across all 404 genes. We then compared the centrality scores for the CRGs to the null distribution to determine those that were significant ($> 5\%$ of the tail of the distribution). Influencers of the network were calculated using the collective influence algorithm available

online (http://github.com/ronammar/collective_influence; accessed June 2019)¹⁸. Enrichment for CRGs and influencers was calculated using a Fisher's exact one-tailed test.

Anthracycline signature and breast cancer CRG network enrichment. Although VIPER¹⁹ was originally designed to identify protein activity associated with a specific transcriptional regulatory program or phenotype, in this study, we adapted this approach to evaluate the enrichment of the CRG network and the doxorubicin signature (or that of other chemotherapeutic agents) (t statistics for each gene) to identify CRGs implicated in drug resistance or sensitivity in breast cancer cell lines. To account for the correlation structure in the expression network, we included the null distribution from 1,000 resampling labels obtained from the drug differential expression analysis in the VIPER model as a parameter. By evaluating the enrichment of mRNAs that were upregulated or downregulated in the anthracycline response signature among genes in the chromatin regulatory network, we identified a consensus list of 38 CRGs. For each CRG, we computed the combined P value across all datasets by using the function combine.test (Fisher's method from the package survcomp)⁵⁴. To test for enrichment of the 12 hits in common from the in vitro and clinical analyses, for each dataset and drug, we computed the VIPER enrichment P value for each CRG and drug signature and evaluated the enrichment of $-\log_{10}(P$ values) and the 12 genes using gene set enrichment analysis from the fgsea package⁵⁵.

Outcome association analyses. We used a Cox proportional hazards model to study the interaction between the expression of each CRG in the breast tumor and the patient's outcome (overall survival) under a specified drug condition in the adjuvant-treated breast cancer meta-cohort. We compared the association between CRG expression and patient outcome under the following sets of drug conditions: (1) anthracycline-treated patients versus non-anthracycline-treated patients (including patients who received non-anthracycline chemotherapy, only endocrine therapy or no therapy); (2) anthracycline-treated patients versus CMF-treated patients; and (3) anthracycline-treated patients versus taxane-treated patients (treated with taxanes alone or in combination with other non-anthracycline agents). The model was adjusted for age, T stage, lymph node status (positive or negative), cohort, MKI67 expression and ER, PR and HER2 status (with the exception of the stratified clinical analysis, where ER, PR and HER2 status was removed accordingly). Hormone therapy was included for ER-positive samples. In HER2-positive tumors, trastuzumab treatment was not included as a covariate because it was not reported. We used the maxstat algorithm⁵⁶ from the survminer package (<https://cran.r-project.org/web/packages/survminer/index.html>) to obtain the optimal threshold to divide profiles with high and low expression for visualization in the Kaplan–Meier plots. For comparing the contrast and Cox proportional hazards probability plots, we defined 'high' as 1 s.d. above the median and 'low' as 1 s.d. below the median. The rms (<https://cran.r-project.org/web/packages/rms/index.html>) and survival (<https://cran.r-project.org/web/packages/survival/index.html>) packages were used for outcome analysis. The survcomp⁵⁴ package (<https://bioconductor.org/packages/release/bioc/html/survcomp.html>) was used for Kaplan–Meier visualization and summary statistics.

In addition to analyzing the full cohort (adjusted for ER, PR and HER2 status), we performed several subset analyses in each of the three clinical subtypes: ER-positive/HER2-negative, HER2-positive and triple-negative breast cancer tumors. Although the subset analysis stratified by ER and HER2 status was not significant ($P < 0.05$), likely owing to the small sample size and lack of treatment information in the HER2-positive cohort, the trends were consistent with the finding that CRGs are predictive of anthracycline benefit irrespective of subgroup. Indeed, for most genes, the directionality of the association was the same, with the exception that one gene changed direction in ER-positive/HER2-negative tumors (HDAC9), two genes changed direction in HER2-positive tumors (NCAPG and CCNA2) and two genes changed direction in triple-negative tumors (NEK11 and NCAPG).

Cell culture. The HCC1954 cell line was obtained from the American Type Culture Collection (ATCC) and maintained as recommended in RPMI medium with 10% FBS at 37 °C in 5% CO₂. Cells were limited to under 20 passages for all experiments. HCC1954 cells are ER negative/HER2 positive but not TOP2A amplified, have a doubling time of ~40 h and are readily transduced with lentiviral particles. Additionally, this cell line is sensitive to doxorubicin and paclitaxel, which is important for showing that KDM4B modulation specifically leads to changes in anthracycline response and not generally to changes in response to small-molecule inhibitors. These observations indicated that it might be a useful model for studying the relationship between response to therapeutics and CRGs.

Lentiviral preparation and infection. Lentiviruses were produced from Lenti-X HEK293T cells via spinfection with polyethylenimine transfection. HEK293T cells were cultured using standard conditions in DMEM containing 10% FBS. HEK293T cells were transfected with polyethylenimine using TRIPz lentiviral vectors, as described below, and cotransfected with packaging vectors psPAX2 and pMD2.G, as previously described⁵⁷. Viral supernatant was collected, ultracentrifuged and used for spinfection.

TRIPz shRNA plasmids. shRNAs in the pTRIPz backbone were obtained from Dharmacon. Clone numbers were as follows: shKDM4B-1, V3THS_316478; shKDM4B-2, V3THS_316479; shKDM4B-3, V3THS_316481; shSCRAM, RHS4743; shKAT6B-88, V3THS_349688; and shKAT6B-92, V3THS_349692. Cells were selected in 2 µg ml⁻¹ puromycin beginning 48 h after transduction and kept under selection for experiments.

Inhibitor dose response. Cells were grown for 3 d in the presence or absence of doxycycline (1.5 µg ml⁻¹) to induce shRNA expression. Cells were then trypsinized, counted and plated in 96-well plates at 3,000–4,000 cells per well with or without doxycycline. Cells were allowed to attach to the well for 24 h, and then the medium was replaced and appropriate concentrations of doxorubicin (AK Scientific, 69518), etoposide (Sigma Aldrich, E1383) or paclitaxel (Fisher Scientific, P3456) or DMSO vehicle control were added. Cells were grown with inhibitors for 48 h when CellTiter-Blue was added to all wells at 5% (vol/vol). Plates were developed at 37 °C and 5% CO₂ overnight. Medium and supernatant was then transferred to 386-well clear-bottom black plates and read on a Flex Station 3 (Molecular Devices) with excitation of 540 nm and emission of 595 nm. CellTiter-Blue emission readings for each treatment well were normalized to those for the DMSO control, and the data were plotted in R-studio. At the end of the inhibitor resistance assays, DMSO control wells were used to calculate growth rate changes upon induction of shRNAs, as the ratio of induced to non-induced cells for both shRNAs to KDM4B and the nonspecific scrambled shRNA. The detailed protocol is available at <https://doi.org/10.17504/protocols.io.k5zcy76>.

Western blots and antibodies. Nuclear or whole-cell extracts were prepared, and protein concentrations were measured. Fifteen micrograms of protein was run on a Bis-Tris gel, transferred at 200 mA for 2 h and blocked in 5% nonfat milk for 1 h. Blots were incubated with primary antibodies overnight and then washed three times in TBST. Blots were then incubated with secondary antibodies with conjugated fluorophores activated at 680 nm and 800 nm (LI-COR; 1:25,000 dilution) for 45 min at room temperature, washed three times and developed on an Odyssey CLx Imaging System (LI-COR). The primary antibodies used included antibodies to KDM4B/MJD2B (Bethyl Laboratories, A301-478A; 1:1,000 dilution), TOP2A (Santa Cruz Biotechnology, sc-3659, clone F-12; 1:1,000 dilution), TOP2B (Santa Cruz Biotechnology, H-286, sc-13059; 1:1,000 dilution), MDR1/ABCB1 (Cell Signaling Technologies, 12683; 1:1,000 dilution), RNF2 (Cell Signaling Technologies, 5694; 1:1,000 dilution) and H3 (Epicyphe, SKU, 13-0001; 1:5,000 dilution). We were unable to validate any of the available antibodies to KAT6B. LI-COR software (Image Studio, version 5.2.5) was used for the following: image adjustment was limited to exposure (brightness and contrast; all lanes done equally and simultaneously) followed by cropping and exporting in .png format. No additional image manipulation was performed.

TOP2 cross-linking. Cells were treated with doxycycline for 4 d to induce shRNA expression as described. Cells were trypsinized and counted, and 1 × 10⁶ cells were resuspended in complete medium. Cells (1 × 10⁶) for input were saved on ice and then processed in RIPA buffer A and RIPA buffer with 150 mM NaCl alongside the etoposide-treated samples. Cells were treated with etoposide (100 µM) for 0, 15 or 60 min with rocking at room temperature. Cells were then washed twice with PBS and resuspended in RIPA buffer A for 10 min. Nuclei were spun down and resuspended in RIPA buffer with 500 mM NaCl for 15 min on ice. The suspension was spun down, and the supernatant was removed. The insoluble chromatin pellet was then solubilized with RIPA containing 1% SDS and Benzonase (1:200) for 10 min at room temperature followed by 20 min on ice. Equal volumes of chromatin fractions were run on western blots as described above. The detailed protocol is available at <https://dx.doi.org/10.17504/protocols.io.2rn9d5e>.

Quantitative RT-PCR. RNA was extracted from cells using Trisure (Bioline), and cDNA was synthesized from 1 µg of RNA using SensiFAST SYBR Lo-Rox (Bioline). Samples were run on a QuantStudio 6 Flex system (Life Technologies), and 2^{-ΔΔCT} was calculated as described by Livak and Schmittgen⁵⁸, where:

$$-\Delta\Delta CT = \frac{(CT_{GOI} - CT_{GAPDH})_{shKAT6B-induced}}{(CT_{GOI} - CT_{GAPDH})_{shKAT6Bnon-induced}}$$

Primers for qPCR were as follows:

KAT6B-BF: AGTGACACCCATCTGTTTG
KAT6B-BR: CCACATGCCCTTTTGGCATTCT
KAT6B-CF: AGGATGTCGTTACTGAAGAGGA
KAT6B-CR: CCCACTCTCACACTCTATTTC
GAPDH-F: GCCAGCCGAGCCACAT
GAPDH-R: CTTTACCAGAGTTAAAGACGACCC

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All data used in this study are available publicly. Clinical cohorts are available under the following GEO identifiers: GSE20685, GSE45255, GSE65194, GSE3494

and GSE1456. Cell line molecular and dose–response data are available from Costello et al.⁶ and the PharmacGX³⁸ package.

Code availability

The code used for the analysis in this study was implemented using R v3.5.3 and is publicly available from the following repository: <https://github.com/cancersysbio/chromatinAntra>.

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Author contributions

J.A.S. and C.C. designed and performed the data analysis. J.G.K. and G.R.C. designed and performed the experiments. J.L.C.-J. provided clinical expertise. G.R.C. and C.C. conceptualized the study. All authors contributed to data interpretation. J.A.S., J.G.K., J.L.C.-J. and C.C. wrote the manuscript. All authors read and approved the final manuscript.

Competing interests

G.R.C. is the founder and a stockholder of Foghorn Therapeutics. C.C. is a scientific advisor to GRAIL and reports stock options as well as consulting for GRAIL and Genentech. J.A.S., J.G.K. and J.L.C.-J. declare no competing interests.

Additional information

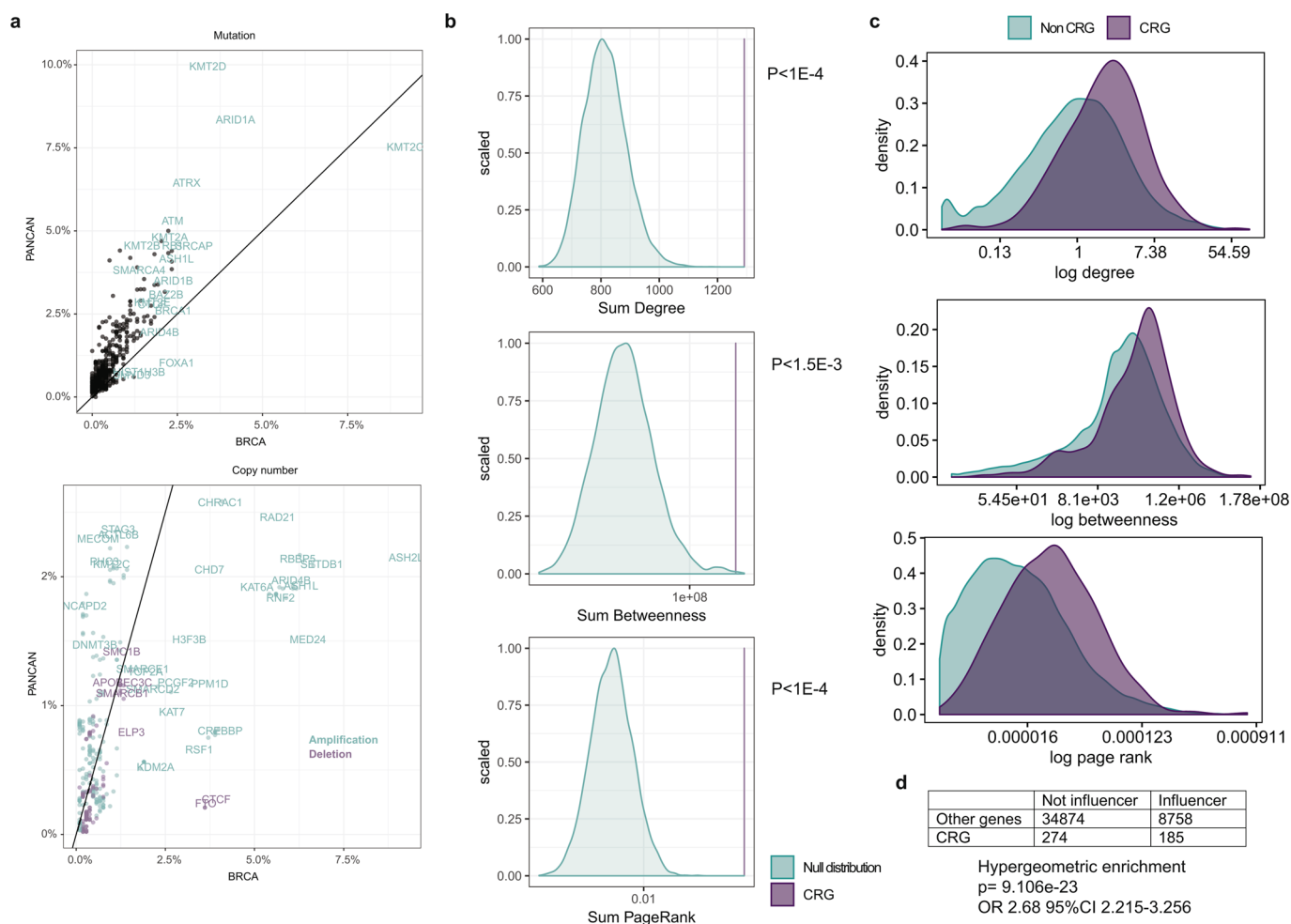
Extended data is available for this paper at <https://doi.org/10.1038/s41591-019-0638-5>.

Supplementary information is available for this paper at <https://doi.org/10.1038/s41591-019-0638-5>.

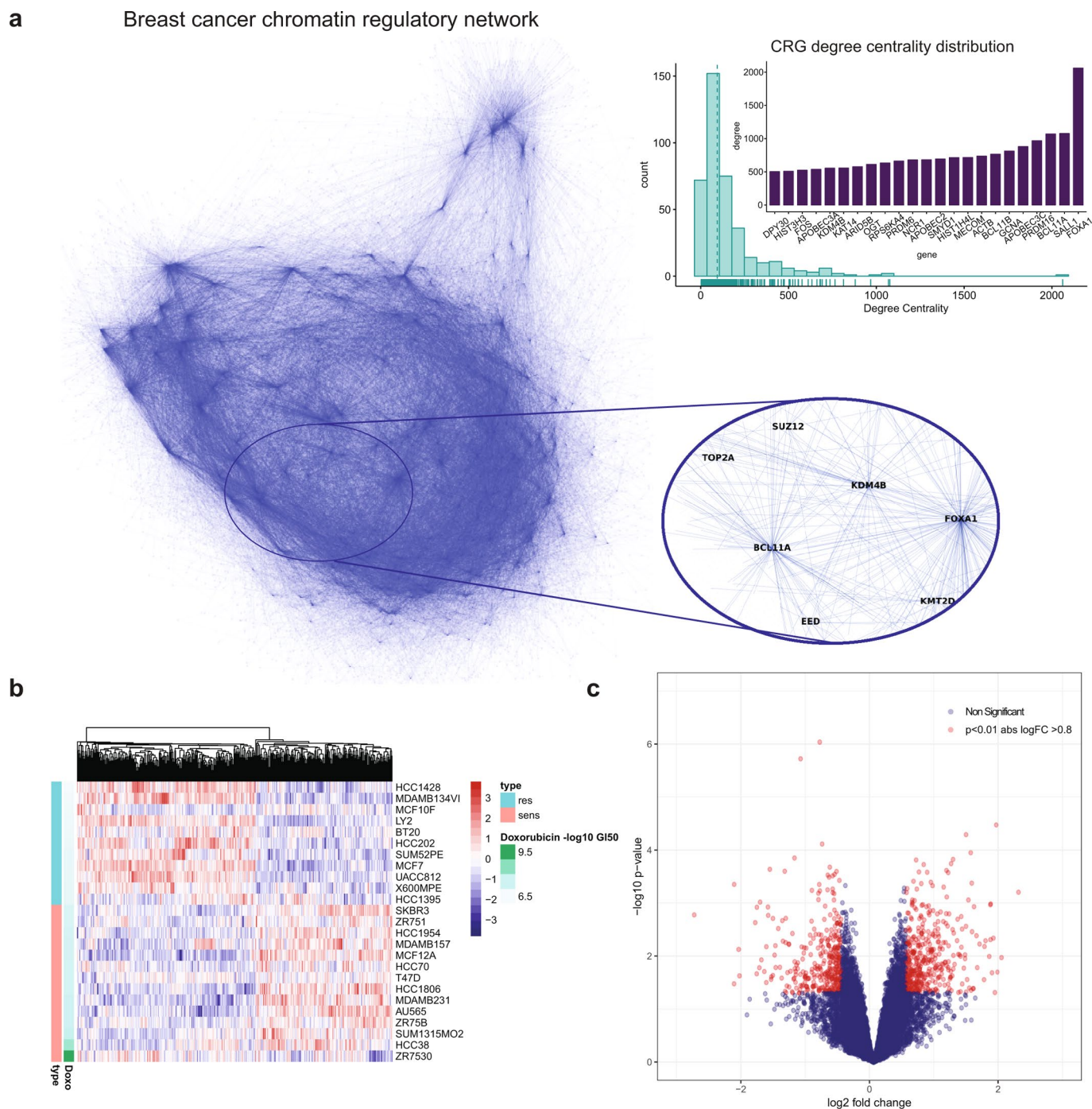
Correspondence and requests for materials should be addressed to G.R.C. or C.C.

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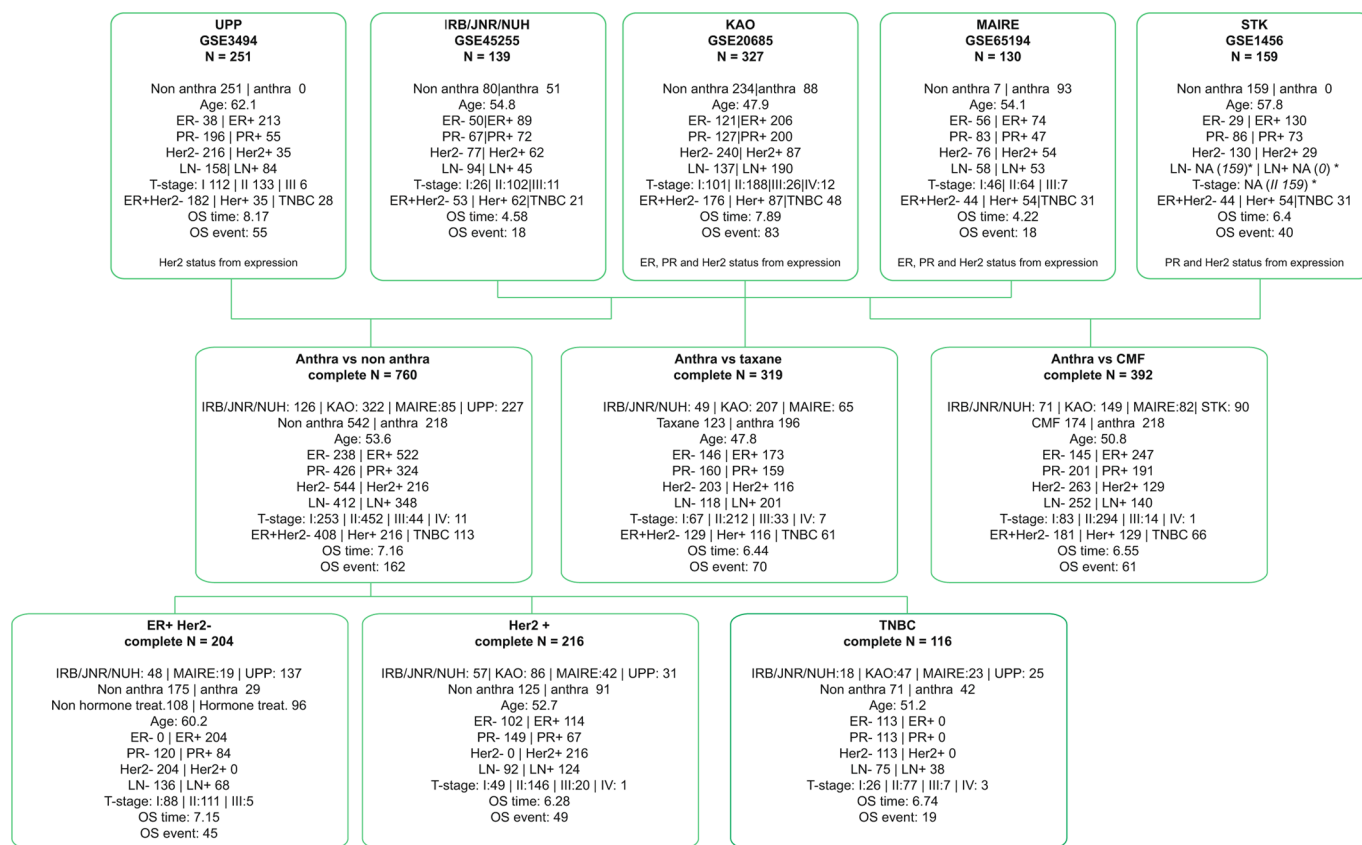
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Extended Data Fig. 1 | Distribution of somatic mutations and copy number alterations and network centrality measures of chromatin regulatory genes. Panel A: Comparison of somatic alterations in chromatin regulatory genes (CRGs) in breast cancer versus pan-cancer. Top panel: Frequency of mutations in CRGs in breast cancer (BRCA, x-axis) versus pancancer (pancan; 32 cancer types excluding breast) based on non-synonymous mutations in whole exome sequencing data from TCGA. Bottom panel: Frequency of copy number amplification and deletion events in CRGs in breast cancer versus pancancer based on TCGA SNP array data. Panel B: Centrality of CRGs. Genome-wide breast cancer network generated from the TCGA breast cancer RNA-seq dataset using ARACNE. Centrality measures were computed for each gene. A null distribution of centrality scores was generated using 10,000 combinations of 404 non-CRGs by summing their degree, betweenness and page rank values (non CRGs) compared with the sum of degree, betweenness and page rank of the CRGs. The comparison between the null distribution and CRG centrality scores highlights the central role of CRGs in the breast cancer network. Panel C: Distribution of network centrality measures among CRGs and non CRGs. Panel D: CRGs are significantly enriched among ‘influencers’ as compared to other genes (Methods).



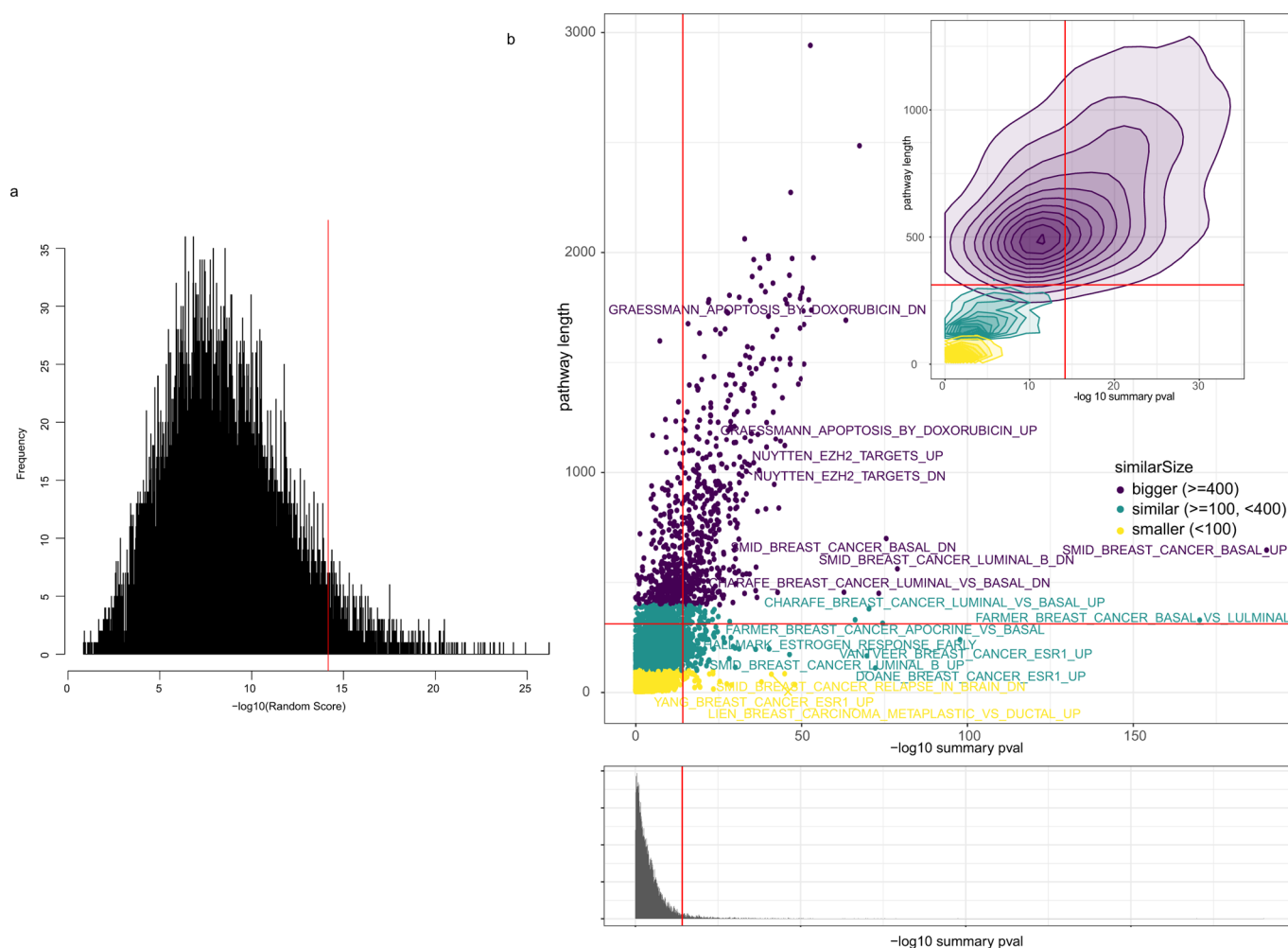
Extended Data Fig. 2 | Breast cancer chromatin regulatory network and doxorubicin response signature. Panel A: Derivation of a breast cancer chromatin regulatory network: A breast cancer chromatin regulatory network was derived from TCGA RNA-seq data using the ARACNE algorithm (Methods). The inset panel shows a region mapping to chromatin regulatory genes (CRGs) with high degree centrality. Each hub of the network is a CRG, while the terminal nodes correspond to their target genes. The histogram and barplot show the degree centrality for all CRGs, where three of the largest network hubs correspond to *FOXA1*, *KDM4B* and *BCL11A*. Panel B: Doxorubicin signature in breast cancer cell lines from Heiser microarray dataset (N=46 independent cell lines). Heatmap of gene expression profiles among resistant (1/3) and sensitive (1/3) breast cancer cell lines where cell lines are sorted by GI50. Panel C: Volcano plot of genes differentially expressed (moderate T-statistic from *limma*) between sensitive versus resistant cell lines.



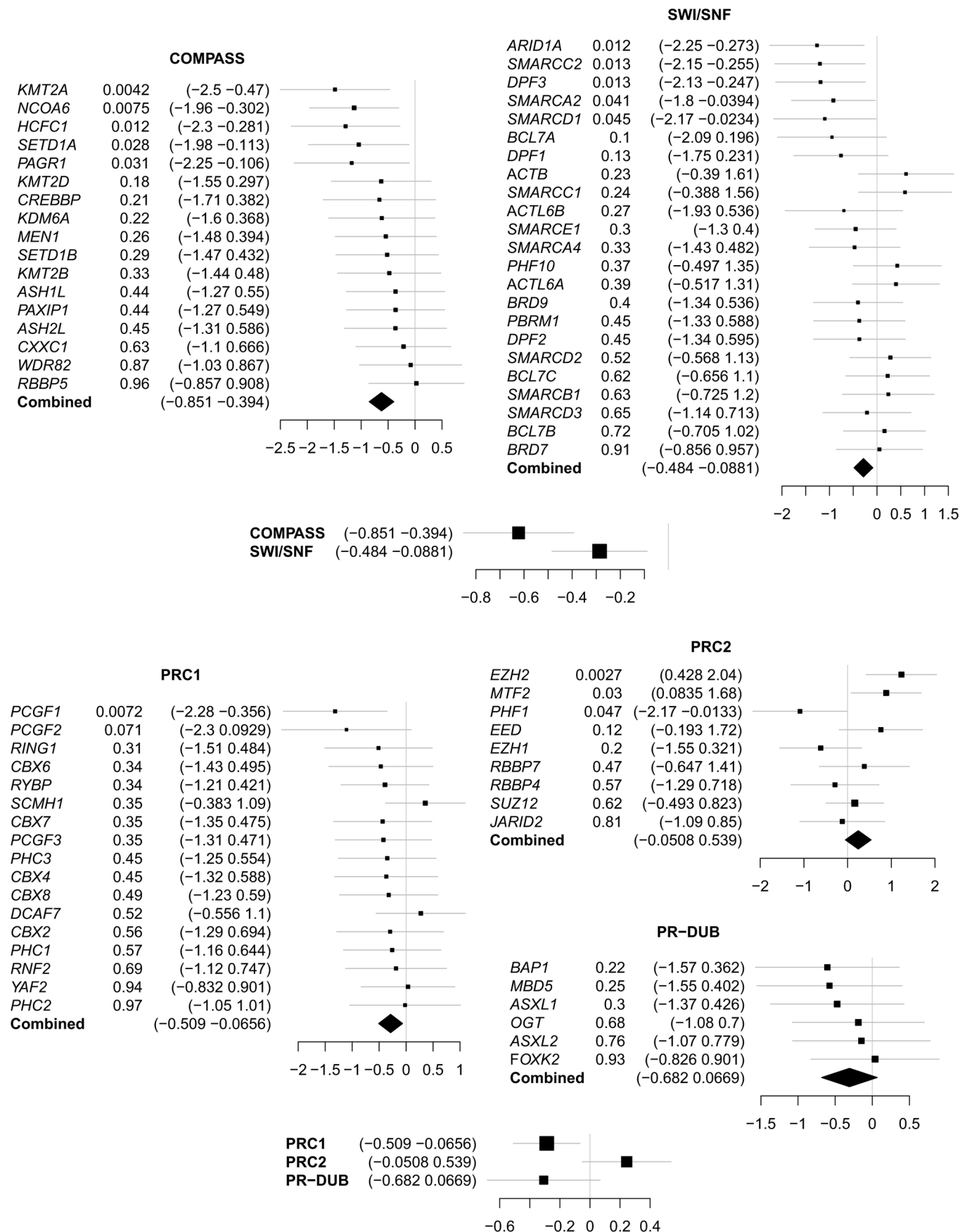
Extended Data Fig. 3 | Summary of adjuvant breast cancer metacohort (N=1006). Top panels describe the 5 individual cohorts used to build the metacohort: UPP (GSE3494), IRB/JNR/NUH (GSE45255), KAO (GSE20685), MAIRE (GSE65194) and STK (GSE1456). All cohorts were profiled on Affymetrix gene expression arrays. Center panels describe the metacohort comparing anthracycline versus non-anthracycline (no therapy, hormone therapy or other chemotherapy), anthracycline versus taxane or anthracycline versus CMF. Bottom panels describe the metacohort stratified by the breast cancer clinical subgroups, namely ER+/HER2-, HER2+ and ER-/HER2- (triple negative). In the ER+/HER2- subgroup, hormonal therapy was included a covariate. *The STK cohort does not report size, t-stage, n-stage or lymph node status, but in the original paper (PMC1410752) the reported mean size is 22 mm and 62% samples are <21 mm, so t-stage=2 was inferred for all samples. Similarly, 38% of samples are lymph node positive but are unmatched to individuals so we treated all samples as LN-.



Extended Data Fig. 4 | Chromatin regulatory genes associated with anthracycline response in the metacohort (a total of 760 individuals treated with anthracycline versus a non-anthracycline-containing regimen were included in this analysis). The forest plots show the log-2 hazard (boxes) of overall survival (OS) based on the expression level of chromatin regulatory genes (CRGs) where error bars represent 95% confidence intervals. P-values (unadjusted for multiple hypothesis testing) are based on the Cox Proportional Hazards model of the interaction between drug and gene expression, adjusted by hormone receptor status, HER2 status, tumor size, lymph node status, *MKI67* expression (from the array data) and cohort. Genes shown in bold were also significant in the in vitro analysis.

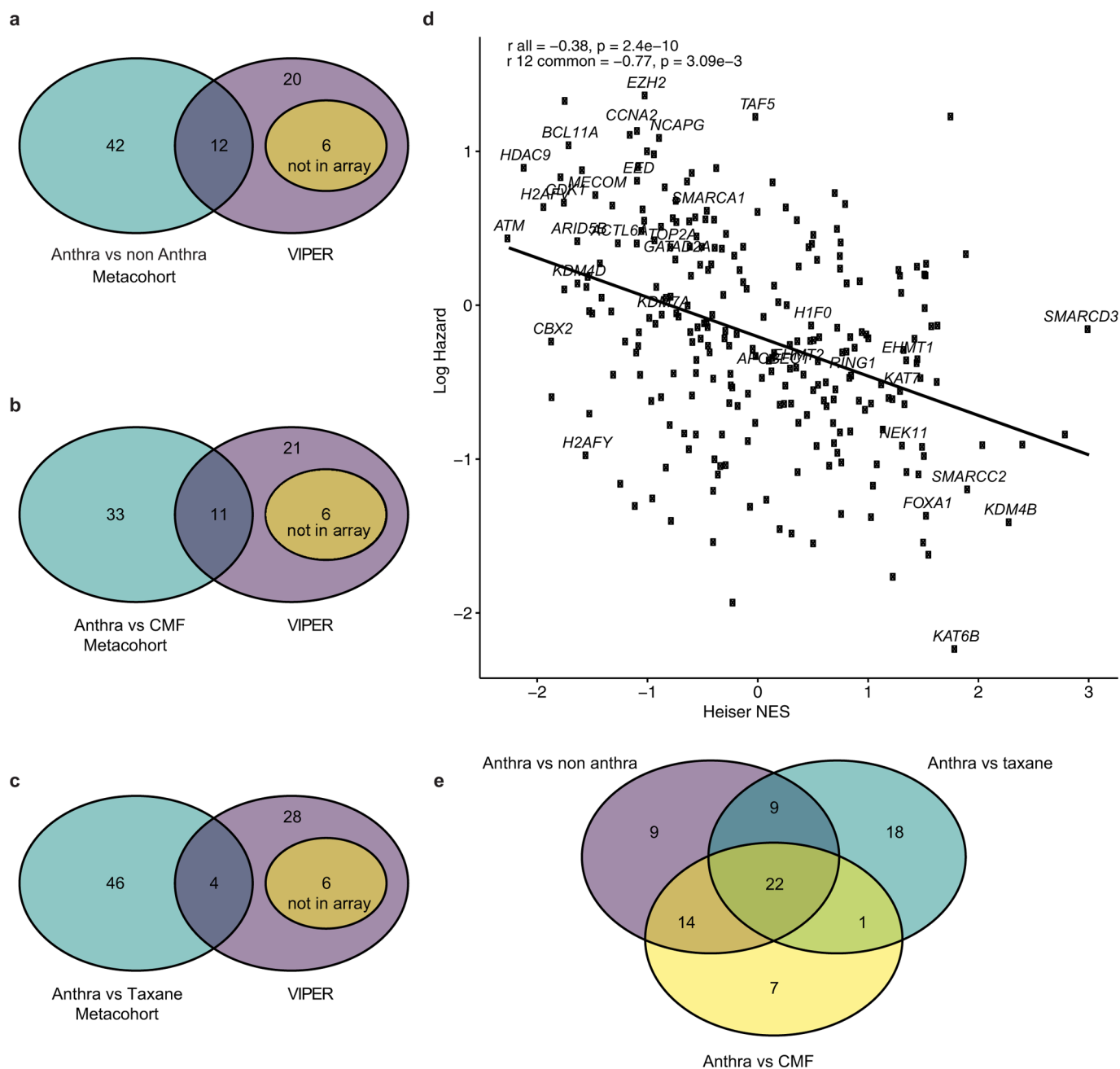


Extended Data Fig. 5 | Enrichment of chromatin regulatory genes among genes associated with anthracycline response. Panel A: Histogram of the $-\log_{10}$ p-values of the association of random gene sets, each consisting of 312 genes (of the 404 CRGs, 312 are present in the microarray data) with overall survival. Of 10,000 gene sets, only 5.93% were more strongly associated with overall survival than the CRGs ($p = 6.8\text{E-}15$, red line). Panel B: Scatter plot (upper panel) of the $-\log_{10}$ p-values of the association between MSigDB gene sets with overall survival (x-axis) and the number of genes in the set (y-axis) (larger signatures display inflated association). Signatures known to be related to breast cancer or chromatin regulation are labeled. The size of the CRG set the p-value for the association with overall survival are indicated by red lines in the density plot in the upper right corner: the association between the CRG set and overall survival is greater than would be expected for gene sets of similar size. Among all signatures, only 4.16% were more strongly associated with overall survival than the CRGs (red line in bottom panel histogram). Both panels A and B display p-values for the Cox Proportional Hazards model of the interaction between treatment and gene expression adjusted by clinical covariates in the anthracycline versus non-anthracycline cohort ($N=760$; see Methods).

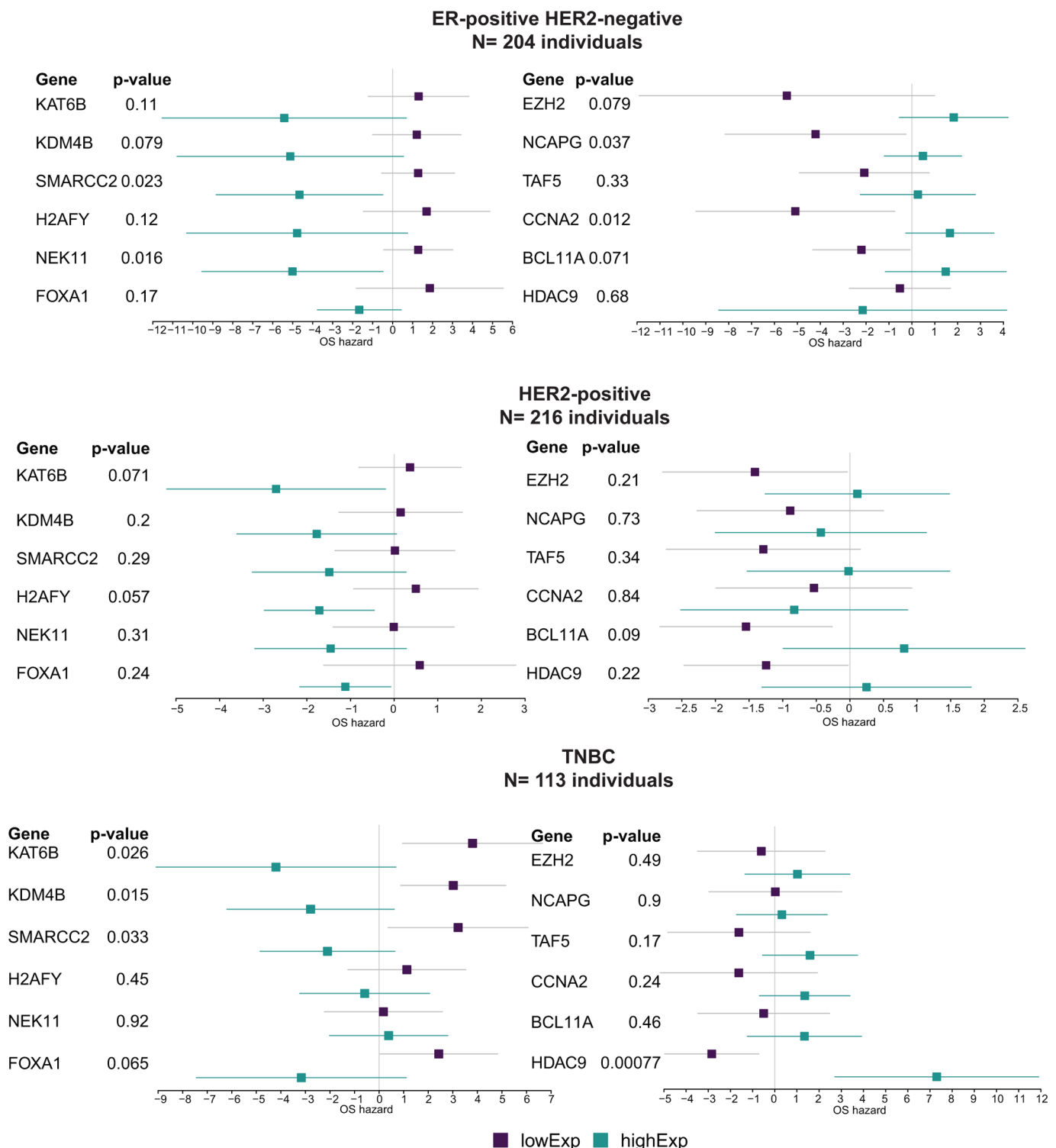


Extended Data Fig. 6 | See next page for caption.

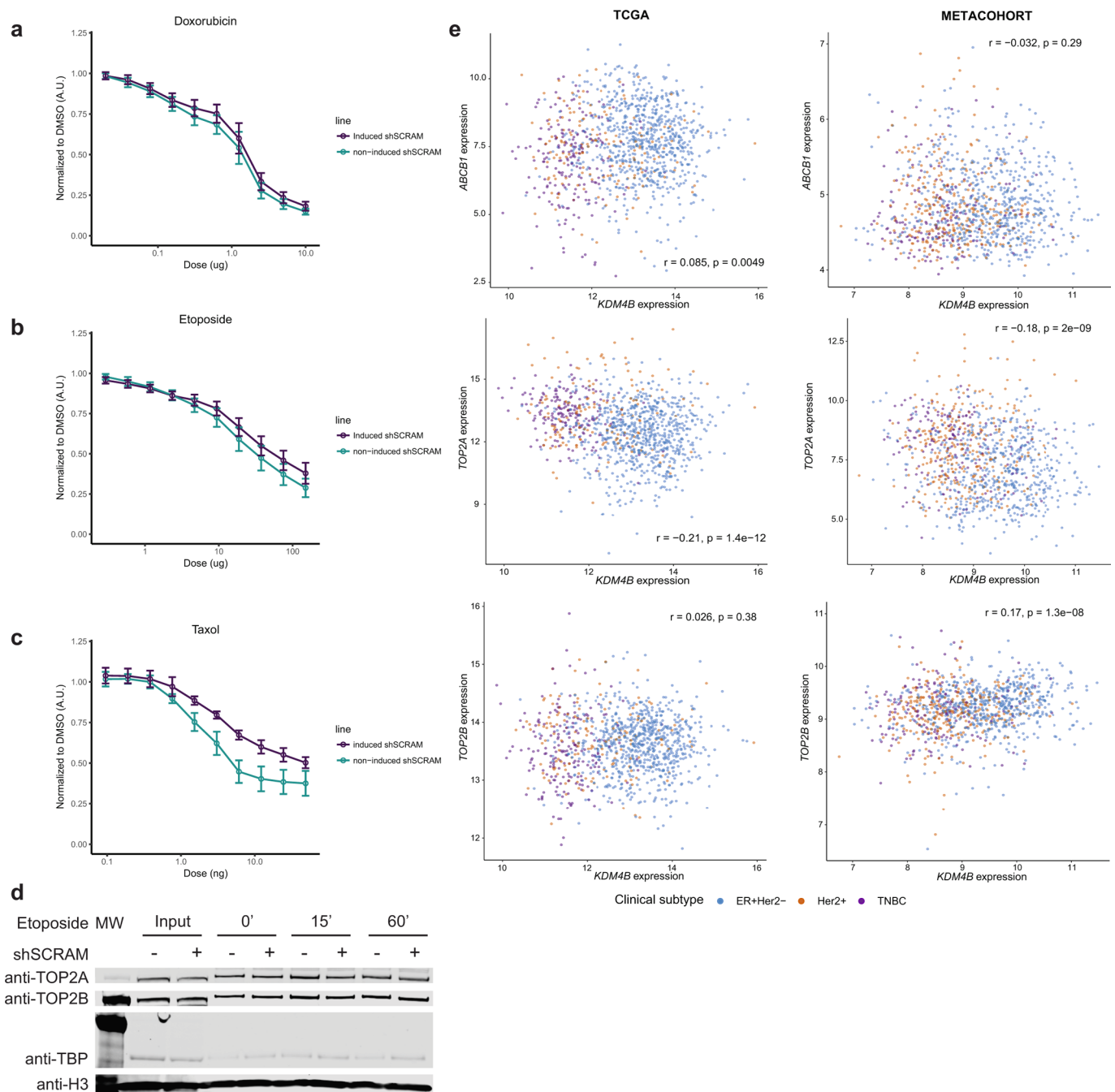
Extended Data Fig. 6 | Chromatin regulatory complexes are associated with anthracycline response. Hazard ratio of overall survival (OS) for members of trithorax (Compass and SWI/SNF) and polycomb (PRC1, PRC2 and PR-DUB) complexes, and the summary statistic (summary p-value from the Cox Proportional Hazard of the interaction between drug and gene expression) for each of the combined complex among anthracycline versus non-anthracycline-treated patients (N=760 individuals). The forest plots show the log-2 hazard ratios (boxes) and error bars represent 95% confidence intervals.



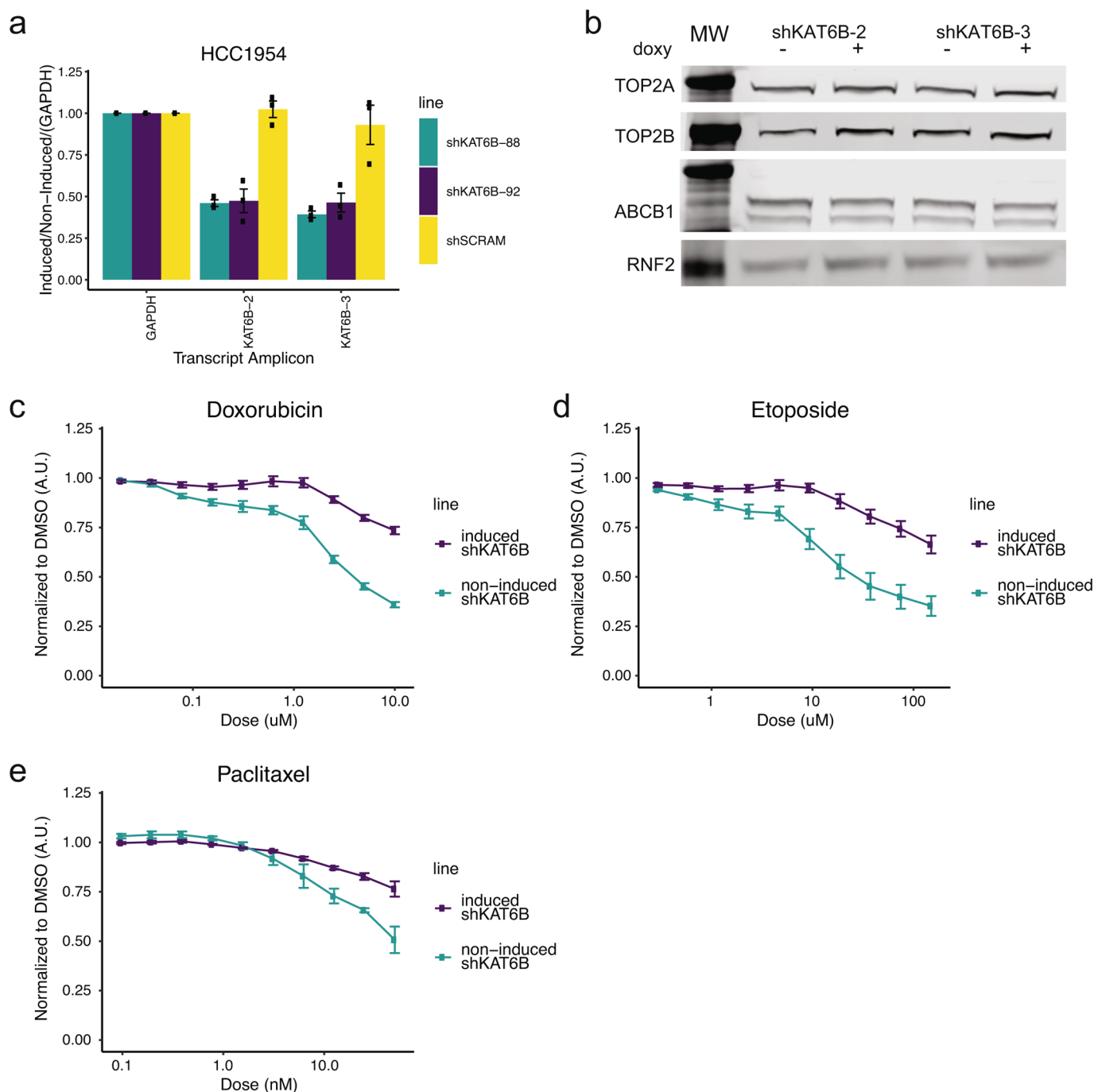
Extended Data Fig. 7 | Comparison of genes associated with anthracycline response in the clinical and in vitro analyses. Panels A, B, C: Venn diagrams indicate the overlap between CRGs associated with anthracycline response in vivo based on the Cox Proportional Hazard model in the breast cancer patient metacohort for different treatment regimens, and in vitro based on VIPER (38 genes). Panel D: Correlation between Heiser microarray dataset Normalized Enriched Score (NES) for each CRG identified via VIPER and the log hazard ratio for overall survival for anthracycline-treated versus non-anthracycline-treated cases in the metacohort (N=760 individual samples). Labels correspond to significant CRGs based on VIPER. Here, r indicates the Pearson correlation across all CRGs and r_{12} indicates the Pearson correlation across the 12 CRGs that were significant in both the clinical and VIPER analyses. Panel E: Venn diagram of the overlap between CRGs that were significant in comparisons of different treatment regimens, namely anthracycline versus non-anthracycline, anthracycline versus CMF, and anthracycline versus taxanes.



Extended Data Fig. 8 | Chromatin regulatory genes associated with anthracycline response stratified by clinical subgroups. Forest plots illustrate the difference in hazards of death (overall survival) among patients treated with anthracyclines versus those who were not treated with anthracyclines from the metacohort across the three clinical subgroups, namely ER+/HER2-, HER2+ and ER-/HER2- (triple-negative, TNBC) patients. P-values indicate the Cox Proportional Hazard interaction model among the two conditions and the expression of the gene, adjusted by age, size, lymph node positive, ER, PR and HER2 status (excluding ER and PR in ER+, HER2 in HER2+ and ER, PR and HER2 in TNBC). The forest plots show the log-2 hazard ratios (boxes) and error bars represent 95% confidence intervals.



Extended Data Fig. 9 | Drug response, accessibility in controls, and transcription correlation in meta-cohort. Panel A: Doxorubicin dose response in scrambled shRNA HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=3$ independent experiments. Center equals mean $\pm$ s.e.m. Panel B: Etoposide dose response in scrambled shRNA HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=3$ independent experiments. Center equals mean $\pm$ s.e.m. Panel C: Paclitaxel dose response in scrambled shRNA HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=3$ independent experiments. Center equals mean $\pm$ s.e.m. Panel D: Western blot of input samples and resolubilized chromatin pellet after wash with 500mM NaCl from HCC1954 cells treated with etoposide for given time, $\pm$ induction of shScramble control. Histone H3 is shown as a loading control. Experiments were repeated independently three times with similar results. See Source Data 2. Panel E: *KDM4B* expression correlations by subtype. Pearson correlation between expression of *KDM4B* and *ABCB1* (top panels), *TOP2A* (center panel) and *TOP2B* (bottom panels) in the TCGA breast cancer cohort (left, $N=1078$ individuals) and the adjuvant breast cancer metacohort (right, $N=760$ individuals) reveals no significant trend.



Extended Data Fig. 10 | Functional evaluation of KAT6B as a mediator of anthracycline response in breast cancer cells. Panel A: RT-qPCR on cDNA showing inducible shRNA knockdown of *KAT6B* with 2 unique shRNA sequences in the HCC1954 breast cancer cell line. $n=3$ independent experiments. Center value equals mean $\pm$ s.e.m. Panel B: Western blot on whole cell extract showing non-induced or induced knockdown shRNA *KAT6B* with no concomitant loss of drug targets TOP2A and TOP2B or gain in the drug efflux pump ABCB1; RNF2 as a loading control. Experiments were repeated independently three times with similar results. See Source Data 3. Panel C: Doxorubicin dose-response curves for *KAT6B* induced shRNAs (shRNA 2, 3) in HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. Panel D: Etoposide dose-response curve for *KAT6B* induced shRNAs (shRNA 2, 3) in HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. Panel E: Paclitaxel dose-response curve for *KAT6B* induced shRNAs (shRNA 2, 3) in HCC1954 cells and non-induced HCC1954 cells normalized to DMSO vehicle. $n=4$; 2 independent experiments of 2 independent shRNAs each for panels c-e. Center value is mean $\pm$ s.e.m.

Reporting Summary

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Statistical parameters

When statistical analyses are reported, confirm that the following items are present in the relevant location (e.g. figure legend, table legend, main text, or Methods section).

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ An indication of whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistics including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☒ ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated
- ☐ ☒ Clearly defined error bars
State explicitly what error bars represent (e.g. SD, SE, CI)

Our web collection on [statistics for biologists](#) may be useful.

Software and code

Policy information about [availability of computer code](#)

Data collection

Public expression datasets have been downloaded using GEOquery package from R and PharmacGX.

Data analysis

All scripts has been developed using R (version 3.5.3). Networks were inferred using Aracne-AP (<https://github.com/califano-lab/ARACNe-AP>) and RTN Bioconductor package (<https://bioconductor.org/packages/release/bioc/html/RTN.html>). Network statistics were obtained using the igraph R package. Cell line differential expression analysis was performed with limma R Bioconductor package (<https://bioconductor.org/packages/release/bioc/html/limma.html>). Enrichment of regulons and the doxorubicin signature was performed with the VIPER R Bioconductor package (<https://www.bioconductor.org/packages/release/bioc/html/viper.html>). Metacohort reprocessing was done using the R bioconductor affy package (<https://bioconductor.org/packages/release/bioc/html/affy.html>). Metacohort batch effect correction was performed using COMBAT with sva Bioconductor package (<http://bioconductor.org/packages/release/bioc/html/sva.html>). Survival analysis was done using the survival and rms R packages. Kaplan-Meier plots and summary statistics were produced with survcomp Bioconductor package (<https://bioconductor.org/packages/release/bioc/html/survcomp.html>). Forest plots were produced with forestplot R package. Venn diagrams were produced with VennDiagram R package. All scripts will be made available at publication. Western blots were cropped in Licor Software: Image Studio Version 5.2.5

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers upon request. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

TCGA and breast cancer cell line data was used for figure 2. Metacohort data was used for figures 3, 4 and 5.

TCGA data was downloaded from gdc.cancer.gov or gdc.cancer.gov/about-data/publications/pancanatlas

Breast cancer cell line data was obtained from: supplementary information from Costello et al. Nat Biotechnol 32 (2014) and PharmacoGX package.

Metacohort expression data was downloaded from GEO accession ids: KAO(GSE20685), IRB/JNR/ NUH (GSE45255), MAIRE (GSE65194), UPS (GSE3494) and STK (GSE1456)

Field-specific reporting

Please select the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/authors/policies/ReportingSummary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

Differential expression analysis:

From Heiser dataset, 53 breast cancer cell lines, 45 exhibited sensitivity or resistance to doxorubicin. From these, cell lines in the top and bottom tertiles were used for downstream analysis, corresponding to 13 resistant and 12 sensitive cell lines. From GRAY dataset, 32 cell lines with doxorubicin data. From these, 11 resistant and 11 sensitive. From FIMM dataset, 18 cell lines with doxorubicin data (6 resistant, 6 sensitive). From CTRPV2 dataset, 35 cell lines with doxorubicin data (13 resistant and 12 sensitive). From GDSC100 dataset, 50 cell lines with doxorubicin data (17 resistant and 17 sensitive). From gCSI dataset, 27 cell lines with doxorubicin data (9 resistant and 9 sensitive).

Breast cancer patient Metacohort:

The total number of samples is N=1006. Specific outcome analysis were done by subtype stratification or drug treatment as specified in the Methods and figure S3.

In vitro validation assays:

Sample size for each experiment is indicated in the figure legend for each experiment. The sample size was chosen based on previous experience for each experiment to yield high power to detect specific effects. No statistical methods were used to predetermine sample size. Sample sizes were N=3 for each shRNA in drug response assays (Total of N=6 or N=9). Sample sizes were chosen to test multiple shRNAs to the same target transcript 3 times per shRNA. Sample sizes were N=3 for Western blots and performed independently with similar results.

Data exclusions

No data was excluded from analysis

Replication

In vitro validation: All attempts at replication were successful and were included in the study. Detailed Materials and Methods for non-standard in vitro experiments are published on protocols.io to aid in reproduction of our findings. Code deposition was described above.

Randomization

For in vitro experiments in this study a randomized non-specific shRNA was used as a control (shRNA Scramble)

Blinding

Blinding was not performed in this study. No IF/IHC/ICC data is presented. Experiments were quantitative and not qualitative

Reporting for specific materials, systems and methods

Materials & experimental systems

n/a	Involved in the study
☒	☐ Unique biological materials
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

KDM4B/JMJD2B (Bethyl Laboratories A301-478A Lot #2, Dilution 1:1000), TOP2A (Santa Cruz Biotech sc-365916, clone F-12, lot #J0114, Dilution 1:1000) TOP2B (Santa Cruz Biotech H-286, sc-13059, lot #L1409, Dilution 1:1000), MDR1/ABCB1 (Cell Signalling Technologies, 12683S, lot #2, Dilution 1:1000), TBP (Abcam, AB818, Lots #GR31577-2 & GR300917-5, Dilution 1:1000); RNF2 (Cell Signalling Technologies, 5694S, Lot#1 & 3, Dilution 1:1000); H3 (Epiccypher, 13-0001, lot 12346001, Dilution 1:5000)

Validation

KDM4B antibody was validated for Western blot with shRNA knockdown with 3 different shRNAs in this study.
 TOP2A: Etoposide treatment leads to a reduction in the TOP2A band in a Western blot in doi: 10.3390/md16060204; cited in at least 5 papers
 TOP2B: Cited in at least 15 papers. Competitive binding study leads to loss of Top2B band in doi: 10.1186/1472-6769-9-1.
 ABCB1: MDR1/ABCB1 (D3H1Q) Rabbit mAb recognizes endogenous levels of total MDR1 protein of human origin. Has been cited 8 times according to CiteAb: <https://www.citeab.com/antibodies/1472663-12683-mdr1-abcb1-d3h1q-rabbit-mab>
 KAT6B: No antibodies were successfully validated so Western Blots were excluded from this study and information on some manufacturers websites are incorrect.
 RNF2: Validated by targeted degradation of RNF2 in mouse ESCs in an unpublished study in our hands
 H3: According to the manufacturer This antibody has been validated for use in ChIP and Western blot. It has also been validated for specificity by dot blot and histone peptide array and it has been confirmed that this antibody: Is specific for H3 C-terminus. Has no cross-reactivity to any modifications

Eukaryotic cell lines

Policy information about [cell lines](#)

Cell line source(s)

HCC1954: ATCC; Cal51: ATCC; HCC1806: ATCC

Authentication

Cell lines were not authenticated, but were purchased directly from ATCC and used at low passage

Mycoplasma contamination

Cell lines tested negative for mycoplasma contamination

Commonly misidentified lines
(See [ICLAC](#) register)

None of the cell lines used are listed in the ICLAC database